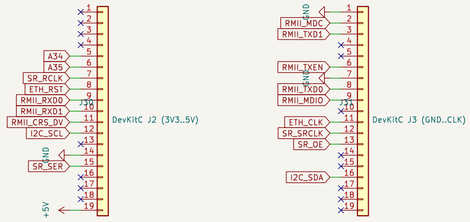
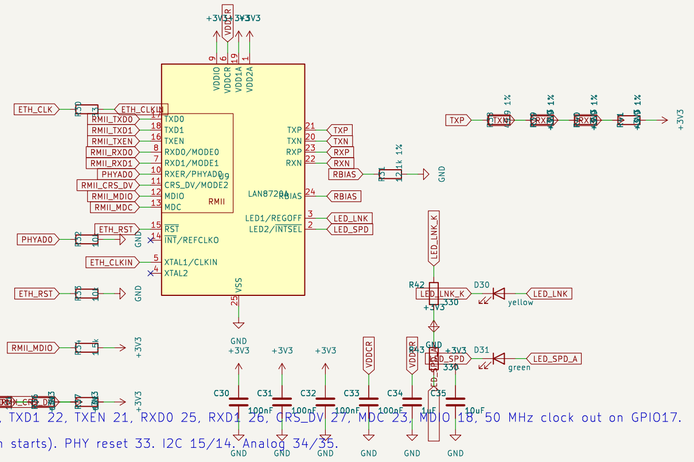


12 V in: reverse protection D25, TVS D26, LM2596 buck to 5 V, AMS1117 to 3.3 V. D28 lets the DevKitC USB power the logic for bench tests.



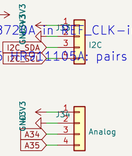


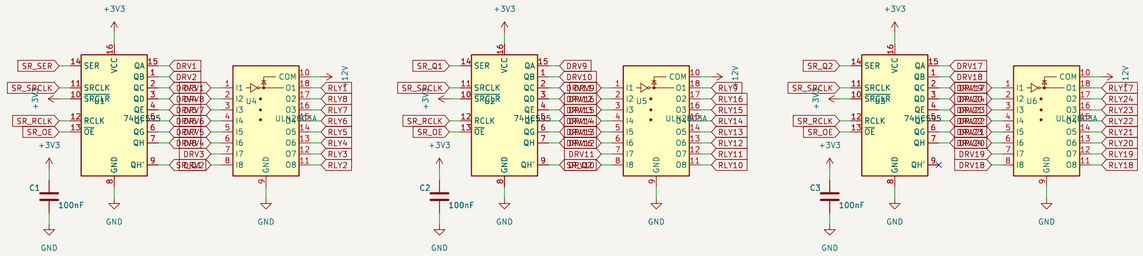
ESP32 DevKitC V4 (38 pins) in two 1x19 sockets. RMII pins are fixed by the ESP32: TXD0 19, TXD1 22, TXEN 21, RXD0 25, RXD1 26, CRS_DV 27, MDC 23, MDIO 18, 50 MHz clock out on GPIO17.

Shift registers: SER 13, SRCLK 16, RCLK 32, OE 4 (pulled up: all relays off until the program starts). PHY reset 33. I2C 15/14. Analog 34/35.

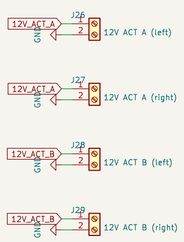
LAN8720A in RMII, CLK-in mode (no crystal): GPIO17 -> R30 -> CLKIN. PHY address 0 (R32). Straps: MODE 111 (R35–R37), REGOFF low (LED1 to GND), nINTSEL high (

RJ45 HR911105A: pairs 1–2 and 3–6, centre taps (4, 5, 7, 8) to 3.3 V with C36/C37, its own LEDs unused (D30/D31 on the board).





Drivers: 74HC595 (3.3 V) -> ULN2803A (coils at 12 V, flyback diodes inside, COM to +12V). One ULN per sheet of 8 relays.



Relays 1–8 (sections 1–8, driver U4)

File: relays_1_8.kicad_sch

Relays 9–16 (sections 9–16, driver U5)

File: relays_9_16.kicad_sch

Relays 17–24 (sections 17–24, driver U6)

File: relays_17_24.kicad_sch

H1 M3 H2 M3 H3 M3

H4 M3 H5 M3 H6 M3

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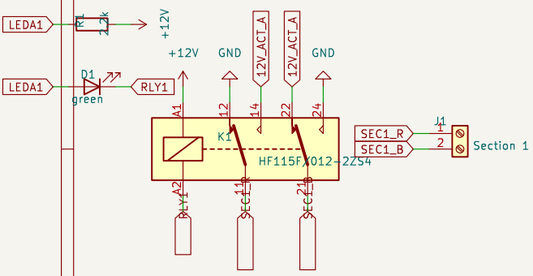
renatorgr

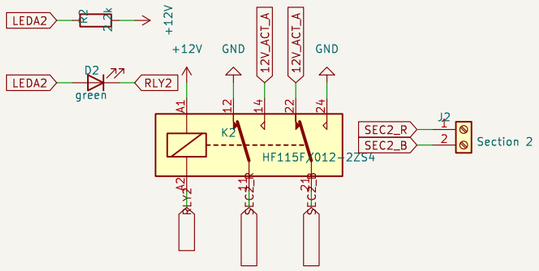
Sheet: /

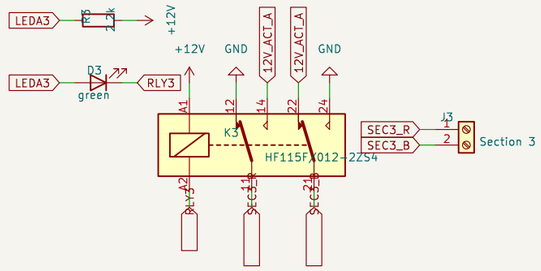
File: Relay24_ESP32.kicad_sch

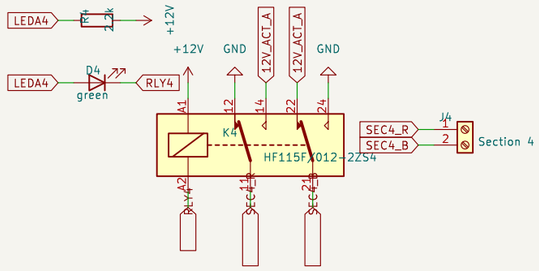
Title: Relay24_ESP32 — 24-section relay board for an ESP32 DevKitC (AgOp

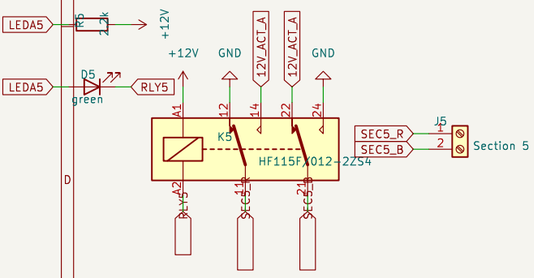
Size: A2	Date: 2026-10-28	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/4

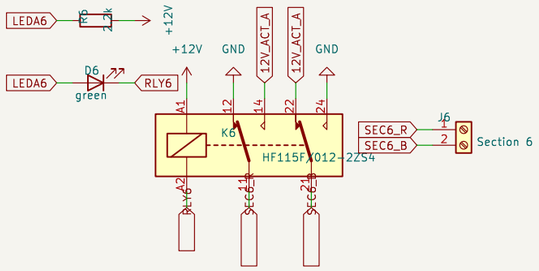


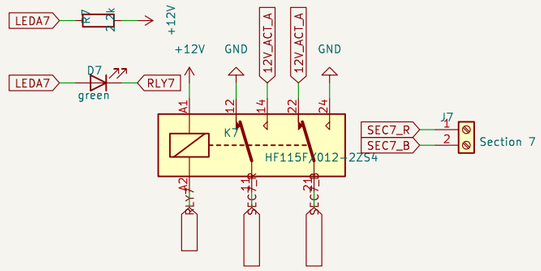


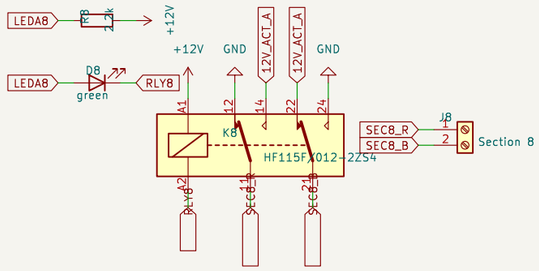












Each section: relay K (DPDT, coil A1 +12V / A2 driven low by the ULN), terminal J (1 = red wire, 2 = black wire). Relay off: red at GND, black at +12V ACT (pull in, closed). Relay on: red +12V ACT, black GND (push out, open).

LED D with R 2.2k across the coil shows the section open. 12V_ACT_A = sections 1–12 (J26/J27), 12V_ACT_B = sections 13–24 (J28/J29).

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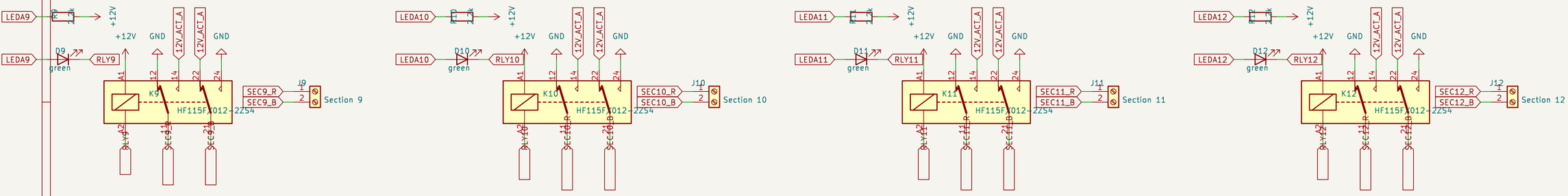
Sheet: /Relays 1–8 (sections 1–8, driver U4)/

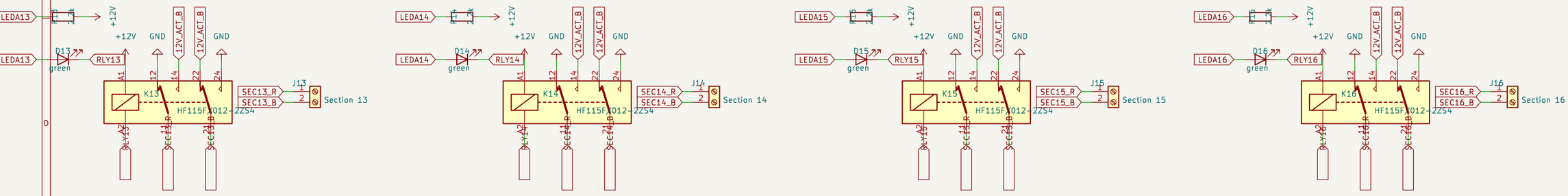
File: relays_1_8.kicad_sch

Title: Relays 1–8 (sections 1–8, driver U4)

Size: A3 | Date: 2026-10-28 | **Rev: 1.0**

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 2/4





Each section: relay K (DPDT, coil A1 +12V / A2 driven low by the ULN), terminal J (1 = red wire, 2 = black wire). Relay off: red at GND, black at +12V ACT (pull in, closed). Relay on: red +12V ACT, black GND (push out, open).

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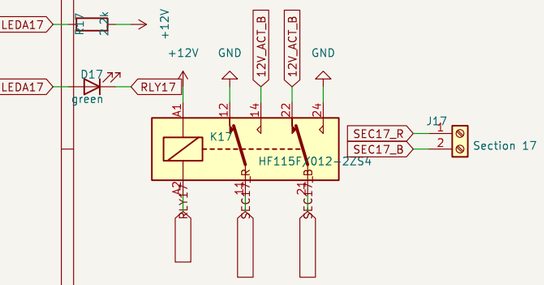
Sheet: /Relays 9–16 (sections 9–16, driver U5)/

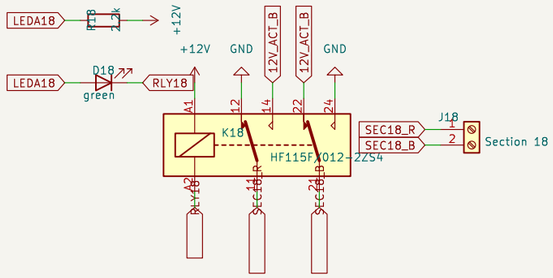
File: relays_9_16.kicad_sch

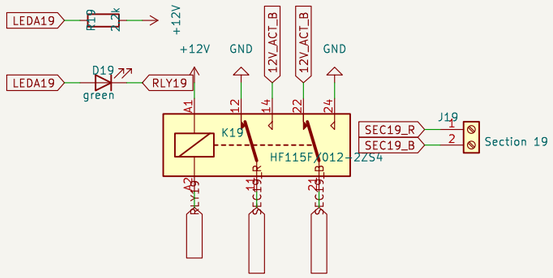
Title: Relays 9–16 (sections 9–16, driver U5)

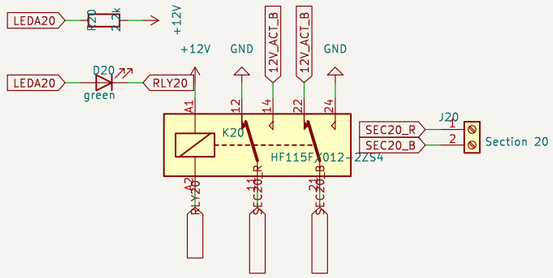
Size: A3 | Date: 2026-10-28 | **Rev: 1.0**

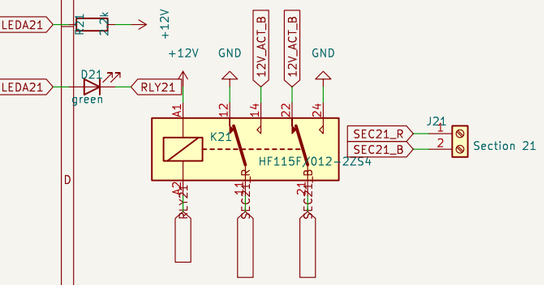
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 3/4

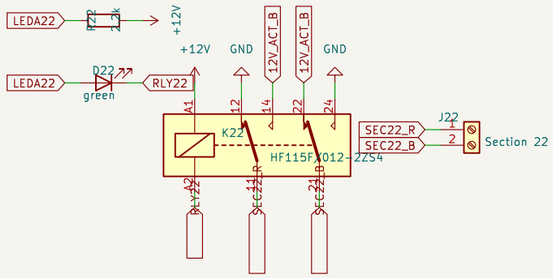


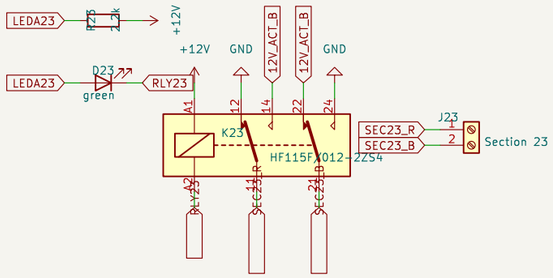


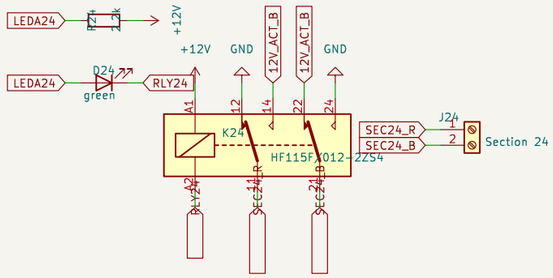












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Sheet: /Relays 17–24 (sections 17–24, driver U6)/

File: relays_17_24.kicad_sch

Title: Relays 17–24 (sections 17–24, driver U6)

Size: A3 | Date: 2026-10-28 | **Rev: 1.0**

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 4/4